Code-Layout, Readability and Reusability

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	5 Marks

This document evaluates the structure, readability and reusability of the codebase.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code (TypeScript and Python)	Yes	Same indentation style used in both frontend and backend
2	Proper File Structure	Files and folders are logically organised	Yes	frontend/{pages,components,services,hooks,utils,layouts,types}; backend/app/{api,services,schemas,db}; tests/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. travelStyle, day_count, itinerary_data
4	Function / Method Names	Functions are descriptively named	Yes	generate_itinerary, build_groq_prompts, verify_supabase_token, deleteTrip
5	Code Comments	Inline and block comments explain logic	Partial	Main parts are commented (e.g. schema.sql); more docstrings can be added
6	Modular Design	Code is split into reusable functions/modules	Yes	Routes, services, schemas and db are in separate layers
7	No Redundant Code	Duplicate or unused code is removed	Partial	A duplicate logo file, an unused ItineraryCreate schema and the OPENAI_API_KEY line in .env.example can be removed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	HTTP errors for AI timeout, rate limit, invalid JSON, DB failure, bad token

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	api.ts (service layer)	TypeScript	Planner, Itinerary and My Trips pages	High
2	useAuth hook / AuthProvider	TypeScript (React)	Layout, auth modal, protected actions	High
3	pdfExport.ts	TypeScript (jsPDF)	Itinerary page and My Trips downloads	Medium
4	itinerary_service.py	Python	Generate route and tests	Medium
5	verify_supabase_token (deps.py)	Python	All /api/trips routes	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4 / 5	Frontend and backend are kept separate
Readability	4 / 5	TypeScript types and clear names
Reusability	4 / 5	Shared hooks, services and dependency functions are reused
Documentation / Comments	3 / 5	README is available; more comments can be added
Overall Score	3.8 / 5	Self rating by the developer